

RV COLLEGE OF ENGINEERING®
DEPARTMENT OF MATHEMATICS
ODD SEMESTER 2024
III SEMESTER, TEST- 1 (Scheme & Solution)
MA231TA

LINEAR ALGEBRA, FOURIER TRANSFORMS AND STATISTICS

Q.No	PART A	Marks
01.	Mean $\bar{x} = \mu'_1 + a = 6$, Variance $\mu_2 = \mu'_2 - \mu_1'^2 = 24$.	2
02.	Standard deviation of $y = 4$, $m_1 = 0.8$, $m_2 = 0.45$ $\tan\theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right = 0.257$ <i>512</i>	2
03.	$(\bar{x}, \bar{y}) = (-2, 1)$	2
04.	$\rho = 1 - \left(\frac{6}{n(n^2-1)} \sum d^2 \right) \Rightarrow \rho = 0.6$.	2
05.	$y = a + bx_1 + cx_2 + dx_3$ The normal equations are $\sum y = na + b \sum x_1 + c \sum x_2 + d \sum x_3$ $\sum x_1 y = a \sum x_1 + b \sum x_1^2 + c \sum x_1 x_2 + d \sum x_1 x_3$ $\sum x_2 y = a \sum x_2 + b \sum x_1 x_2 + c \sum x_2^2 + d \sum x_2 x_3$ $\sum x_3 y = a \sum x_3 + b \sum x_1 x_3 + c \sum x_2 x_3 + d \sum x_3^2$.	2

PART-B

1									4
	x	f	fx	$d = x - \bar{x}$	fd	fd^2	fd^3	fd^4	
								10000	
	5	1	5	-10	-10	100	-1000		
	10	2	20	-5	-10	50	-250	1250	
	12	3	36	-3	-9	27	-81	243	
	14	12	168	-1	-12	12	-12	12	
	15	4	60	0	0	0	0	0	
	19	5	95	4	20	80	320	1280	
	22	3	66	7	21	147	1029	7203	
		30	450		0	416	6	19988	
	Mean $\bar{x} = \frac{450}{30} = 15$, $\mu_0 = 1$, $\mu_1 = 0$, $\mu_2 = 13.87$, $\mu_3 = 0.2$, $\mu_4 = 666.27$. Coefficient of skewness $\beta_1 = \frac{\mu_3}{\mu_2^{\frac{3}{2}}} = 0.0039$ (Positive skewness) coefficient of kurtosis $= \beta_2 = \frac{\mu_4}{\mu_2^2} = 3.46$ (Lepto kurtic)								
									1
									3
									2



DEPARTMENT OF MATHEMATICS

Academic year 2024-2025 (Odd Sem)

(CIE I- B. E. PROGRAMS, EC, EE, EI, ET)

(CIE I- B. E. PROGRAMS, EC, EE, EI, ET)

2b	Seven students got the following percentage of marks in two subjects x and y. Compute their rank correlation coefficient. <table border="1" style="width: 100%; border-collapse: collapse;"><tr><td style="width: 15%;">Marks in x</td><td style="width: 12.5%;">68</td><td style="width: 12.5%;">64</td><td style="width: 12.5%;">75</td><td style="width: 12.5%;">64</td><td style="width: 12.5%;">75</td><td style="width: 12.5%;">40</td><td style="width: 12.5%;">64</td></tr><tr><td>Marks in y</td><td>62</td><td>58</td><td>68</td><td>81</td><td>68</td><td>48</td><td>70</td></tr></table>	Marks in x	68	64	75	64	75	40	64	Marks in y	62	58	68	81	68	48	70	6	L2	2					
Marks in x	68	64	75	64	75	40	64																		
Marks in y	62	58	68	81	68	48	70																		
3	The following table gives the stopping distance y in meters of a motor bike moving at a speed of x Kms/hour when the breaks are applied. <table border="1" style="width: 100%; border-collapse: collapse;"><tr><td style="width: 12.5%;">x</td><td style="width: 12.5%;">16</td><td style="width: 12.5%;">24</td><td style="width: 12.5%;">32</td><td style="width: 12.5%;">40</td><td style="width: 12.5%;">48</td><td style="width: 12.5%;">56</td></tr><tr><td>y</td><td>0.4</td><td>0.7</td><td>1.3</td><td>1.9</td><td>2.7</td><td>3.8</td></tr></table> Find the coefficient of correlation between the speed and stopping distance and estimate the regression line of stopping distance on speed and speed on stopping distance.	x	16	24	32	40	48	56	y	0.4	0.7	1.3	1.9	2.7	3.8	10	L3	3							
x	16	24	32	40	48	56																			
y	0.4	0.7	1.3	1.9	2.7	3.8																			
4	Determine the equation of the regression plane connecting x_1, x_2, and y given <table border="1" style="width: 100%; border-collapse: collapse;"><tr><td style="width: 33%;">Diffusion time(hours) x_1</td><td style="width: 11%;">1.5</td><td style="width: 11%;">2.5</td><td style="width: 11%;">0.5</td><td style="width: 11%;">1.2</td><td style="width: 11%;">2.6</td><td style="width: 11%;">0.3</td></tr><tr><td>Sheet resistance (Ohms-cm) x_2</td><td>66</td><td>87</td><td>69</td><td>141</td><td>93</td><td>105</td></tr><tr><td>Current gain (y)</td><td>5</td><td>8</td><td>7</td><td>9</td><td>10</td><td>9</td></tr></table> Estimate y at $x_1 = 1.8$ and $x_2 = 112$.	Diffusion time(hours) x_1	1.5	2.5	0.5	1.2	2.6	0.3	Sheet resistance (Ohms-cm) x_2	66	87	69	141	93	105	Current gain (y)	5	8	7	9	10	9	10	L3	3
Diffusion time(hours) x_1	1.5	2.5	0.5	1.2	2.6	0.3																			
Sheet resistance (Ohms-cm) x_2	66	87	69	141	93	105																			
Current gain (y)	5	8	7	9	10	9																			
5	Let $\mathbb{R}^2, \mathbb{R}^3, M_{2 \times 2}$ (set of all 2×2 real matrices) be vector spaces with usual addition and scalar multiplication. Verify whether the following sets forms a subspace or not. Justify your answer. (i) $s_1 = \{(x, y, z) \in \mathbb{R}^3 : x - 3y + 4z = 0\}$ (ii) $s_2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$ (iii) $s_3 = \{A \in M_{2 \times 2} : \det(A) = 0\}$ (iv) $s_4 = \{(x, y) \in \mathbb{R}^2 : \text{set of all lines passing through the origin}\}.$	10	L2	2																					

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	4	6	-	-	6	2	2	-	-	-
	Test		4	26	20	-	-	16	34	-	-	-

$$b_{xy} = \frac{n\sum xy - \sum x \sum y}{n\sum y^2 - (\sum y)^2} = 11.4563$$

regression line of stopping distance on speed

$$(y - \bar{y}) = b_{yx}(x - \bar{x}) \Rightarrow y = -1.2342 + 0.5843x$$

regression line of speed on stopping distance

$$(x - \bar{x}) = b_{xy}(y - \bar{y}) \Rightarrow x = 15.3786 + 11.4563y.$$

04

x_1	x_2	y	x_1^2	x_2^2	$x_1 * y$	$x_2 * y$	$x_1 * x_2$
1.5	66	5	2.25	4356	7.5	330	99
2.5	87	8	6.25	7569	20	696	217.5
0.5	69	7	0.25	4761	3.5	483	34.5
1.2	141	9	1.44	19881	10.8	1269	169.2
2.6	93	10	6.76	8649	26	930	241.8
0.3	105	9	0.09	11025	2.7	945	31.5
8.6	561	48	17.04	56241	70.5	4653	793.5

Substituting in Normal equations, we get

$$6\beta_0 + 8.6\beta_1 + 561\beta_2 = 48$$

$$8.6\beta_0 + 17.04\beta_1 + 793.5\beta_2 = 70.5$$

$$561\beta_0 + 793.5\beta_1 + 56241\beta_2 = 4653$$

On solving $\beta_0 = 3.144$, $\beta_1 = 0.462$, $\beta_2 = 0.045$

$$\therefore y = 3.144 + 0.462x_1 + 0.045x_2$$

$$x_1 = 1.8 \text{ and } x_2 = 112$$

$$y = 9.0156.$$

05.

(i) Subspace. Justification.

(ii) Not a subspace. Justification.

(iii) Not a subspace. Justification.

(iv) Subspace. Justification.

2a

$\mu_1 = 0$

$\sigma = 4 \Rightarrow \sigma^2 = \mu_2 = 16$

$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\mu_4}{256}$

Distribution is Leptokurtic if $\beta_2 > 3$

$\Rightarrow \frac{\mu_4}{256} > 3 \Rightarrow \mu_4 > 768$

Distribution is Mesokurtic if $\mu_4 = 768$

Distribution is Platykurtic if $\mu_4 < 768$.

2

1

1

2b

x	y	R_x	R_y	$d = R_x - R_y$	d^2
68	62	3	5	-2	4
64	58	5	6	-1	1
75	68	1.5	3.5	-2	4
64	81	5	1	4	16
75	68	1.5	3.5	2	4
40	48	7	7	0	0
64	70	5	2	3	9
					38

$$\sum d^2 = 38, n = 7, m_1 = 2, m_2 = 3, m_3 = 2$$

$$\text{Rank Correlation } \rho = 1 - \frac{6 \left[\sum d^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \frac{m_3(m_3^2-1)}{12} \right]}{n(n^2-1)} = 0.2678$$

2

3

x	y	xy	x^2	y^2
16	0.4	6.4	256	0.16
24	0.7	16.8	576	0.49
32	1.3	41.6	1024	1.69
40	1.9	76	1600	3.61
48	2.7	129.6	2304	7.29
56	3.8	212.8	3136	14.44
216	10.8	483.2	8896	27.63

$$n = 6, \bar{x} = \frac{\sum x}{n} = 36, \bar{y} = \frac{\sum y}{n} = 1.8,$$

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{\{n \sum x^2 - (\sum x)^2\} \{n \sum y^2 - (\sum y)^2\}}} = 0.9826, b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} = 0.0843,$$

5

3

2